INNOVATION IN MOTION

MOBILITY AND RIDE-HAILING ANALYTICS DOMAIN

UBER PROJECT

Transforming Raw UBER Data into Actionable Insights using Power BI

The Challenge

Uber

- <u>38% Revenue Leakage:</u> **57,000 lost bookings** representing massive untapped revenue potential

- <u>High Cancellation Crisis:</u> **37,000 cancelled rides** indicating systematic operational failures

- <u>Volatile Growth Pattern:</u> Monthly revenue fluctuations ranging from **-12.3%** to **+15.0%,** creating unpredictable business performance

- <u>Driver-Customer Rating Gap:</u> **Driver ratings (4.23)** lag behind **customer satisfaction (4.40)**, suggesting service quality issues

- <u>Incomplete Data Tracking:</u> **29,942 blank cancellation** reasons indicating poor data governance

Business Impact

These challenges are costing Uber millions in lost revenue, damaging brand reputation, and creating competitive vulnerabilities in an aggressive ride-hailing market.

Canva
presentation design

ABOUT ME

Uber

SOUMYADEEP DHAR

SKILLS

Advanced Excel
SQL
Power Bi
Python
ETL

CURRENTLY DOING VIRTUAL INTERNSHIP AT CODEBASICS

FOCUS ON TURNING COMPLEX DATA INTO CLEAR, ACTIONABLE BUSINESS INTELLIGENCE

We'll answer Ten critical questions:
Starting with why are we losing 57,000 bookings? and
How do we recover that 19.7 million rupees?
Then, what's causing 37,000 cancellations?
Where are our peak demand gaps?
And
Why is our driver performance lagging behind customer expectations? and many more.....

BY THE END, WE'LL SEE A PRIORITIZED ROADMAP— IMMEDIATE WINS, SHORT-TERM FIXES, AND LONG-TERM STRATEGY— ALL DRIVEN BY DATA.

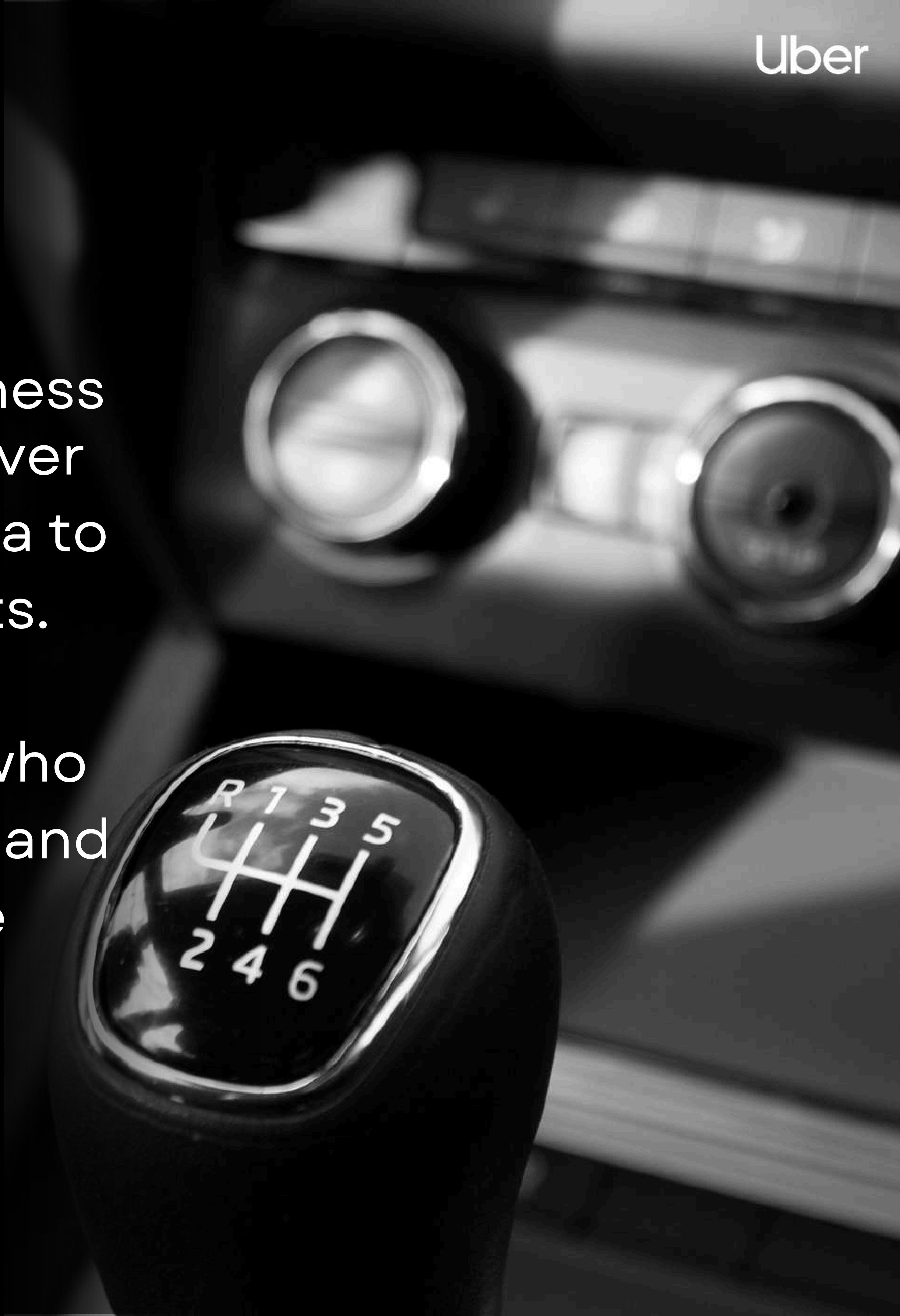
INTRODUCTION

Uber is a global mobilityplatform powered bytechnologythat links drivers and passengers via aneasy-to-use mobile app. Instantridebookings, real-time tracking, transparentpricing, and a seamlesstravelexperience are all made possible by its design,which aims to make transportation accessible, economical, and effective. Uber remains capable of operating on a huge worldwide scale thanks to real-time data and sophisticated analytics. With **171 million** monthly active platform consumers **(MAPCs) in Q4 2024**, the company completed about **11.27 billion trips** globally in 2024, a **14% increase** from the previous year. Also, driver and courier **earnings** reached **US$20 billion,** demonstrating the platform's high level of marketplace activity. It operates in approximately **70 countries** and **15,000 cities** worldwide and heavily relies on data analytics for demand forecasting, dynamic pricing, route optimization, real- time marketplace matching, and safety, as these metrics demonstrate.

MY ROLE

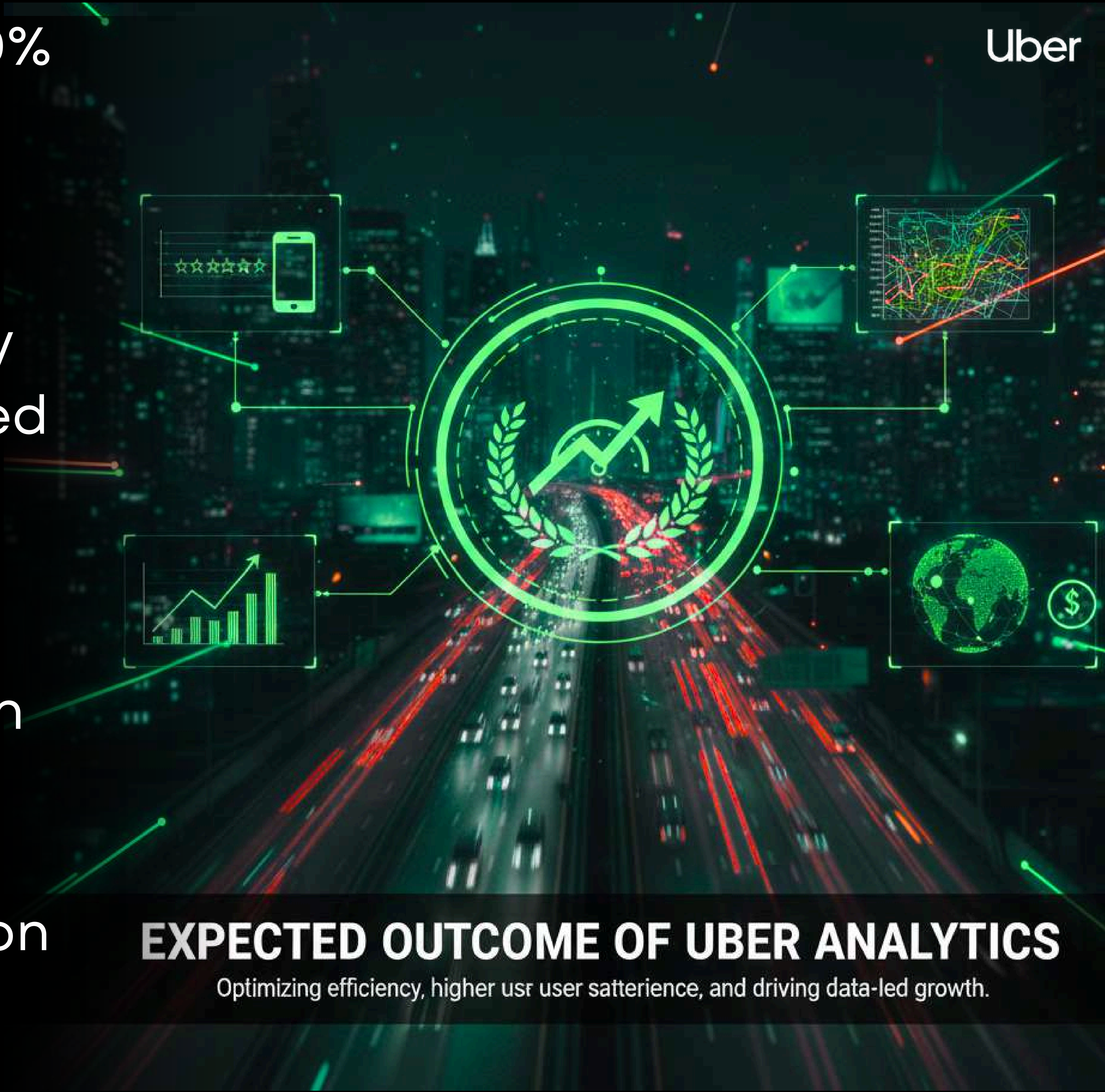
In my role as a data analyst for this Uber Business Intelligence project, I have complete control over the data lifecycle from comprehending the data to providing dashboards with actionable insights.

Additionally, I want to become the detective who finds revenue leaks, uncovers hidden patterns, and makes strategic recommendations that the leadership can actually put into practice.



- Identify root causes of revenue leakage and booking failures
- Uncover operational inefficiencies causing cancellations
- Discover demand patterns to optimize driver allocation
- Analyze customer behavior to improve retention and satisfaction
- Provide data-driven recommendations to increase revenue by 15-20% within 6 months

- Reduce lost bookings by 50% (potential 19% revenue increase)
- Decrease cancellations by 30% (11,000 more completed rides)
- Achieve consistent 3-7% month-over-month growth
- Improve driver ratings to match customer satisfaction levels (4.40+)



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ANALYTICAL APPROACH

Data Collection → Data Cleaning → Data Preparation (Transformation + Reduction)

DATA COLLECTION

Key steps :

Unprocessed data of UBER raw file containing booking id, date ,time etc. various parameters

Source :

Google Drive

DATA CLEANING

Key steps :

Eliminate null values and duplicates.
Fix errors and inconsistencies
Standardizethe formatsfor dates, currencies, andlocations.

Tools use :

MS Excel, Power BI

Power BI

DATA PREPARATION

Key steps :

Data transformation and modeling. Create calculated fields, normalized categorical data.Establish relationships between tables etc.

Visualization :

Variety of Data Visuals

LIVE DASHBOARD OVERVIEW

DATA MODEL

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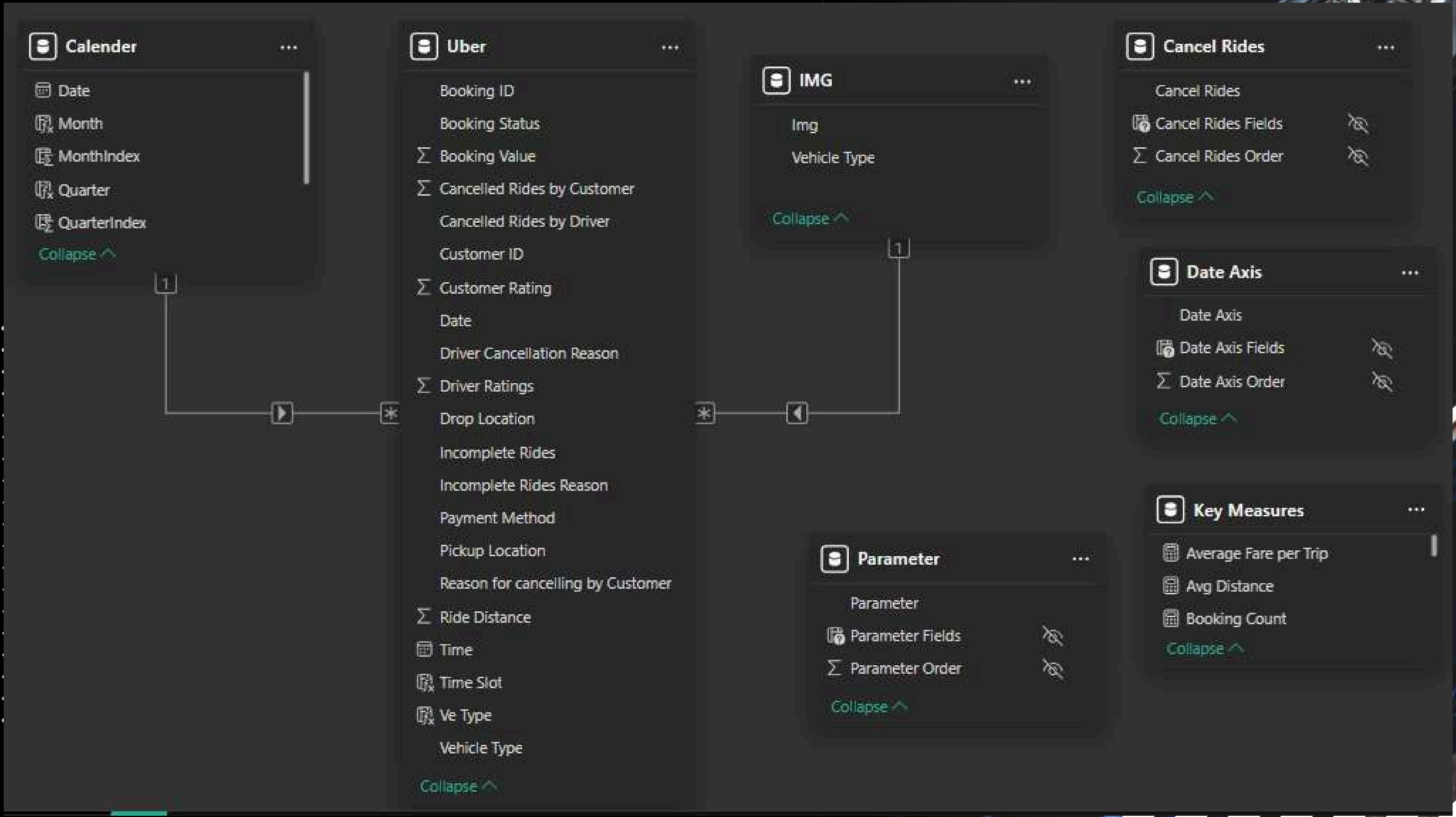


KEY MEASURES

Average Fare per Trip =

DIVIDE([Revenue],[TotalTrips])

Avg Distance = AVERAGE(Uber[Ride Distance])

Booking Count = DISTINCTCOUNT(Uber[Booking ID])

Bookings Remove Status Filter =

CALCULATE([Booking

Count],All(Uber[Booking Status]))

Completed Bookings = CALCULATE([Booking

Count],Uber[Booking Status]="Completed")

Con% =

var vehicle_remove_filter=CALCULATE(SUM(Uber[Booking Value]),ALL(Uber[Ve Type]))

var revenue =SUM(Uber[Booking Value])

return DIVIDE(revenue,vehicle_remove_filter)

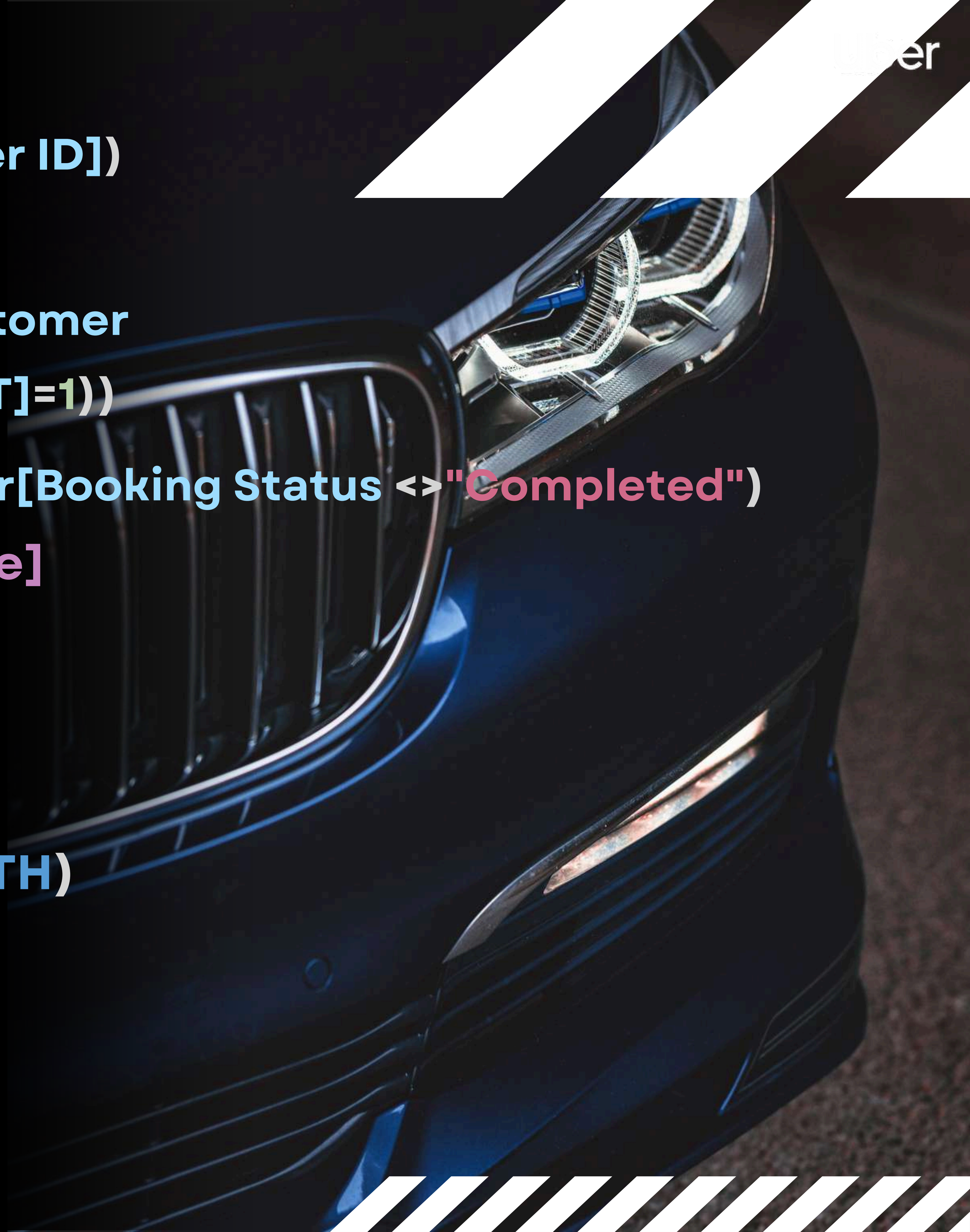
KEY MEASURES

```
Customer Count=DISTINCTCOUNT(Uber[Customer ID])

Firsttime =
COUNTROWS(FILTER(SUMMARIZE(Uber,Uber[Customer
ID],"COUNT",COUNT(Uber[Customer ID])),[COUNT]=1))

Lost Bookings = CALCULATE([Booking Count],Uber[Booking Status <>"Completed")

MoM Growth % = var Current_Month = [Revenue]
var Previous_Month =
    CALCULATE(
        [Revenue],
        DATEADD('Calender'[Date],-1, MONTH)
    )
RETURN
DIVIDE(Current_Month - Previous_Month,
Previous_Month) *100
```



KEY MEASURES

RegularRider=
COUNTROWS(FILTER(SUMMARIZE(Uber,Uber[Customer ID],"COUNT",COUNT(Uber[Customer ID])),[COUNT]>=3))

ReturnRider=
COUNTROWS(FILTER(SUMMARIZE(Uber,Uber[Customer ID],"COUNT",COUNT(Uber[Customer ID])),[COUNT]=2))

Revenue= SUM(Uber[Booking Value])

Top Month Revenue =
MAXX(VALUES(Calender[Month]),[Revenue])

Total Distance = SUM(Uber[Ride Distance])

Total Trips= COUNTROWS(Uber)

Lost Revenue = [Lost Bookings] * [Average Fare per Trip]

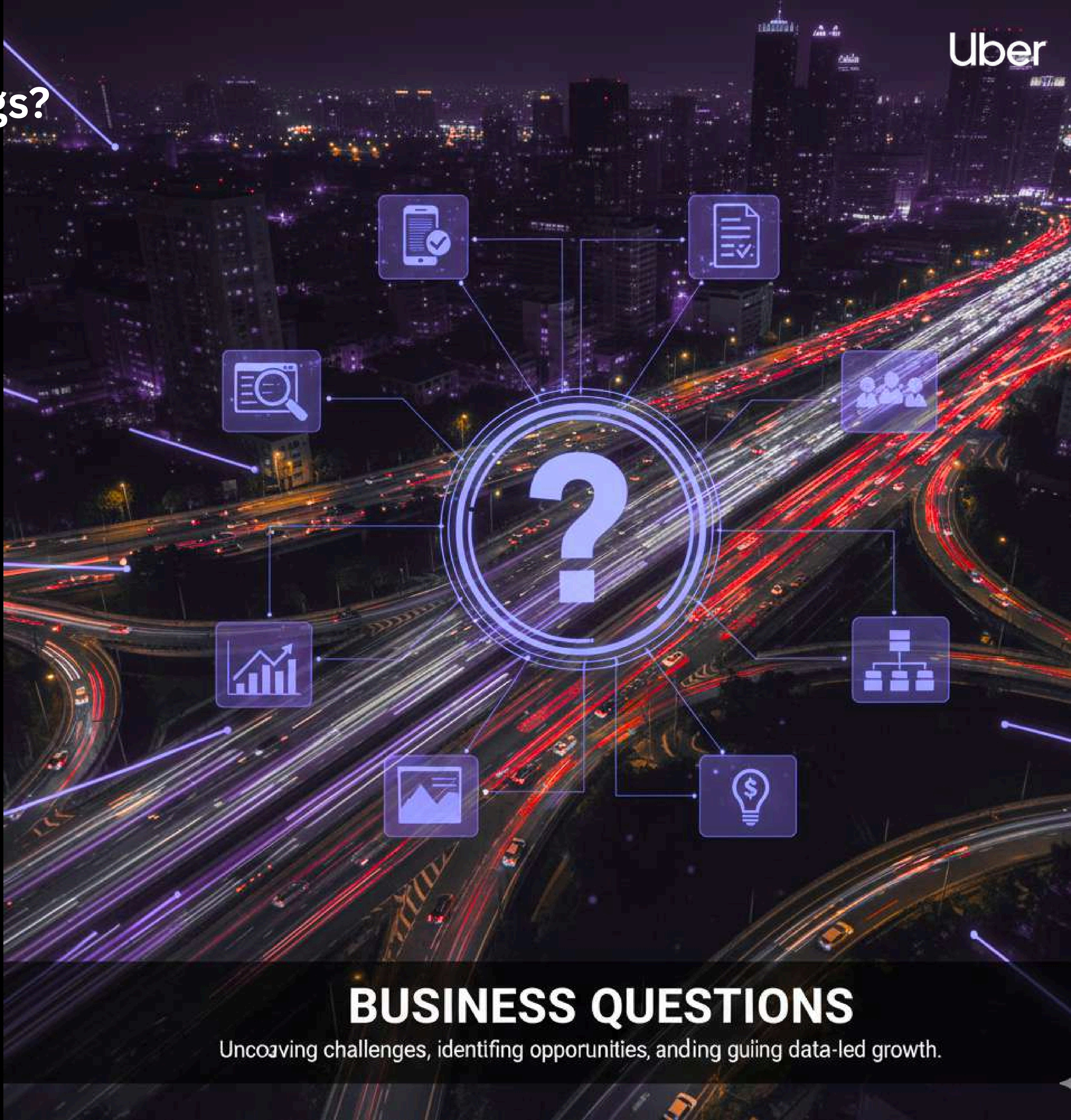
BUSINESS QUESTION 1:

What is the financial impact of lost bookings?

With 57,000 lost bookings at an average fare of ₹345.64, we're looking at approximately 19.7 million rupees in lost revenue potential. That's 38% of revenue.

Even more concerning: These aren't random losses. They're concentrated during peak hours when driver supply doesn't meet demand. That means we're failing customers exactly when they need us most. If we could recover just 50% of these lost bookings through better driver allocation and surge pricing, we'd add nearly 10 million rupees to the bottom line.

Implement surge pricing optimization; driver incentives during peak hours; real-time capacity monitoring



BUSINESS QUESTION 2 :

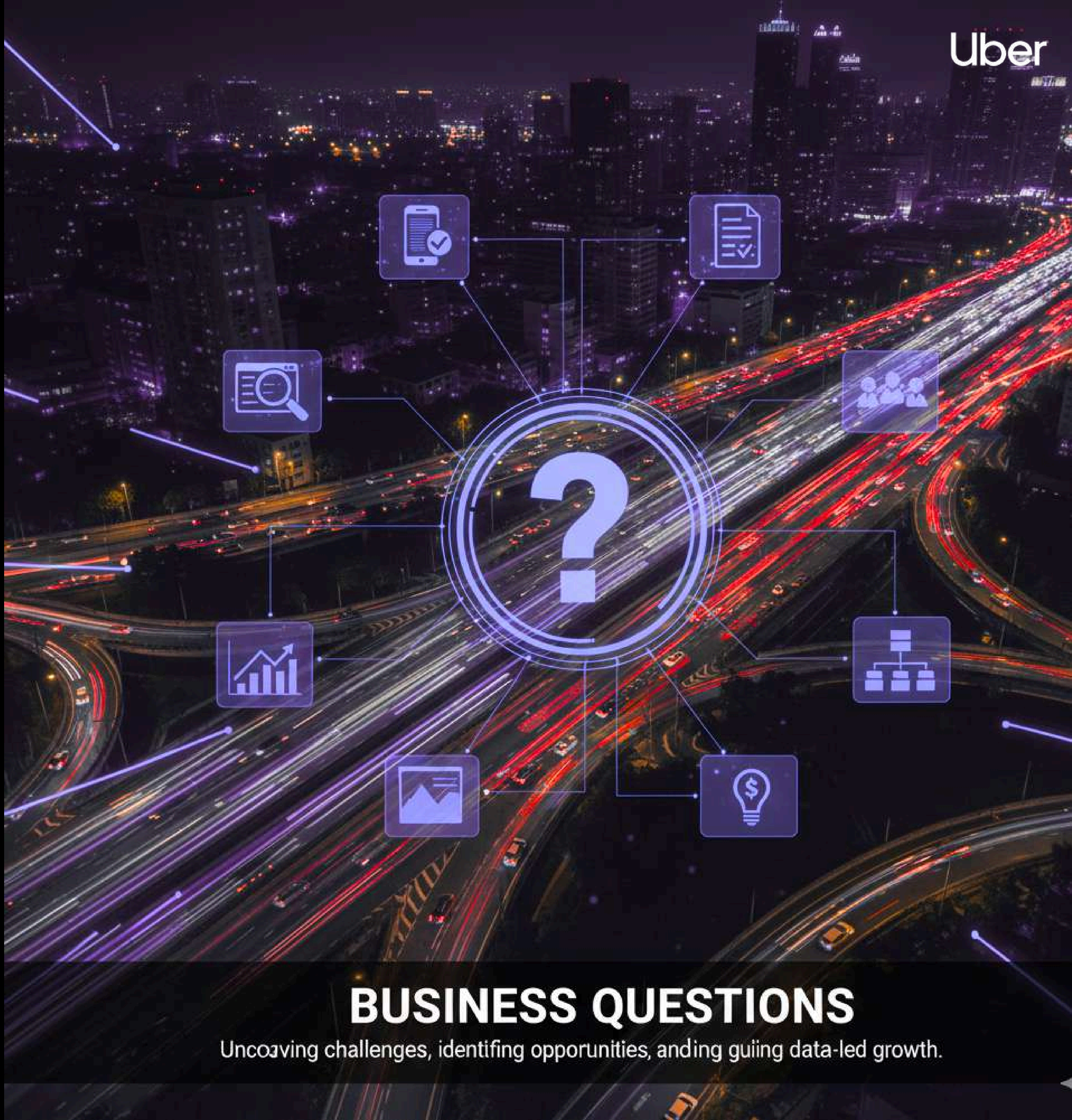
Are there seasonal or temporal patterns affecting performance?

Distance traveled dropped from 232,000 km in January to 197,000 km in February—a sharp 15% decline that repeats through the year.

Month-over-month growth shows wild swings: (-ve) 12% in Feb, (+ve) 15% in March, (-ve) again in Aug and Sept, slight recovery in Oct then declining in Nov and Dec.

These patterns could indicate seasonal factors, competitive pressure, or operational issues that compound during specific months. Without understanding the 'why,' we can't prevent future declines

Investigate underlying causes (competition, weather, events); develop winter strategy with targeted promotions



BUSINESS QUESTION 3:
Where are our peak demand gaps?

Three critical gaps identified from Location Page heatmap:

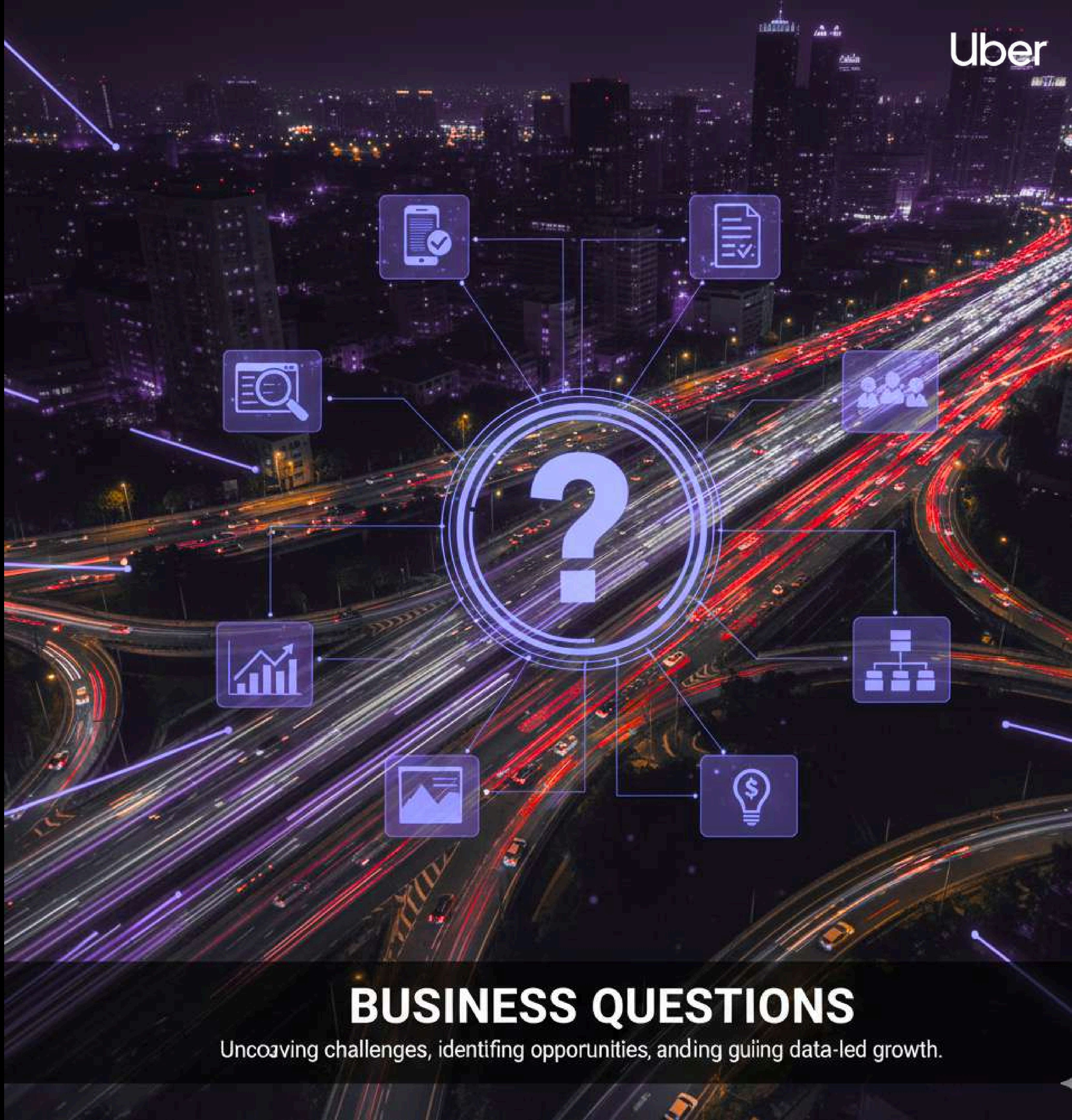
Evening Rush (6-9 PM): 4,796 rides demand, ~1,822 unfulfilled daily

Afternoon Commute (3-6 PM): 4,194 rides demand, ~1,594 unfulfilled daily

Mid-Morning Business (9 AM-12 PM): 3,813 rides demand, ~600 unfulfilled daily

Total Gap: ~4,000 unfulfilled rides daily = ₹1.39M lost per day

Strategic driver placement at hotspots during 6-9 PM; establish dedicated routes



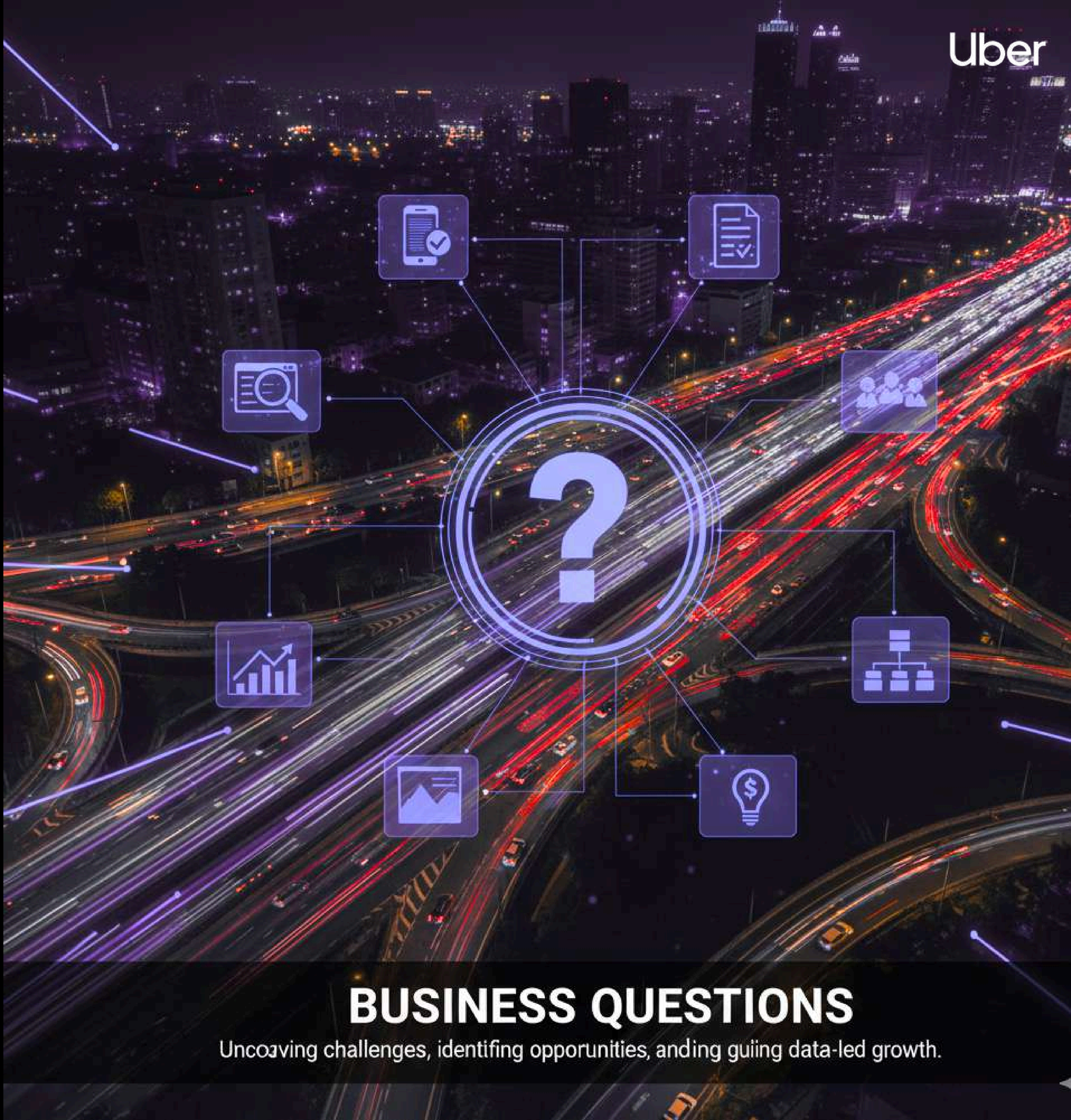
BUSINESS QUESTION 4:
Why are customers cancelling rides?

Out of 37,000 cancelled rides, 81% (29,942) have no recorded cancellation reason.

Of the tracked cancellations, 6,835 were customer-related issues, and several hundred were due to 'more than permitted people' or driver-side problems.

This blank data represents a critical failure in our data governance. Without knowing why customers cancel, we can't fix the underlying issues. It's like trying to cure a disease without diagnosing it first.

Implement mandatory cancellation reason selection; improve driver-rider matching algorithms



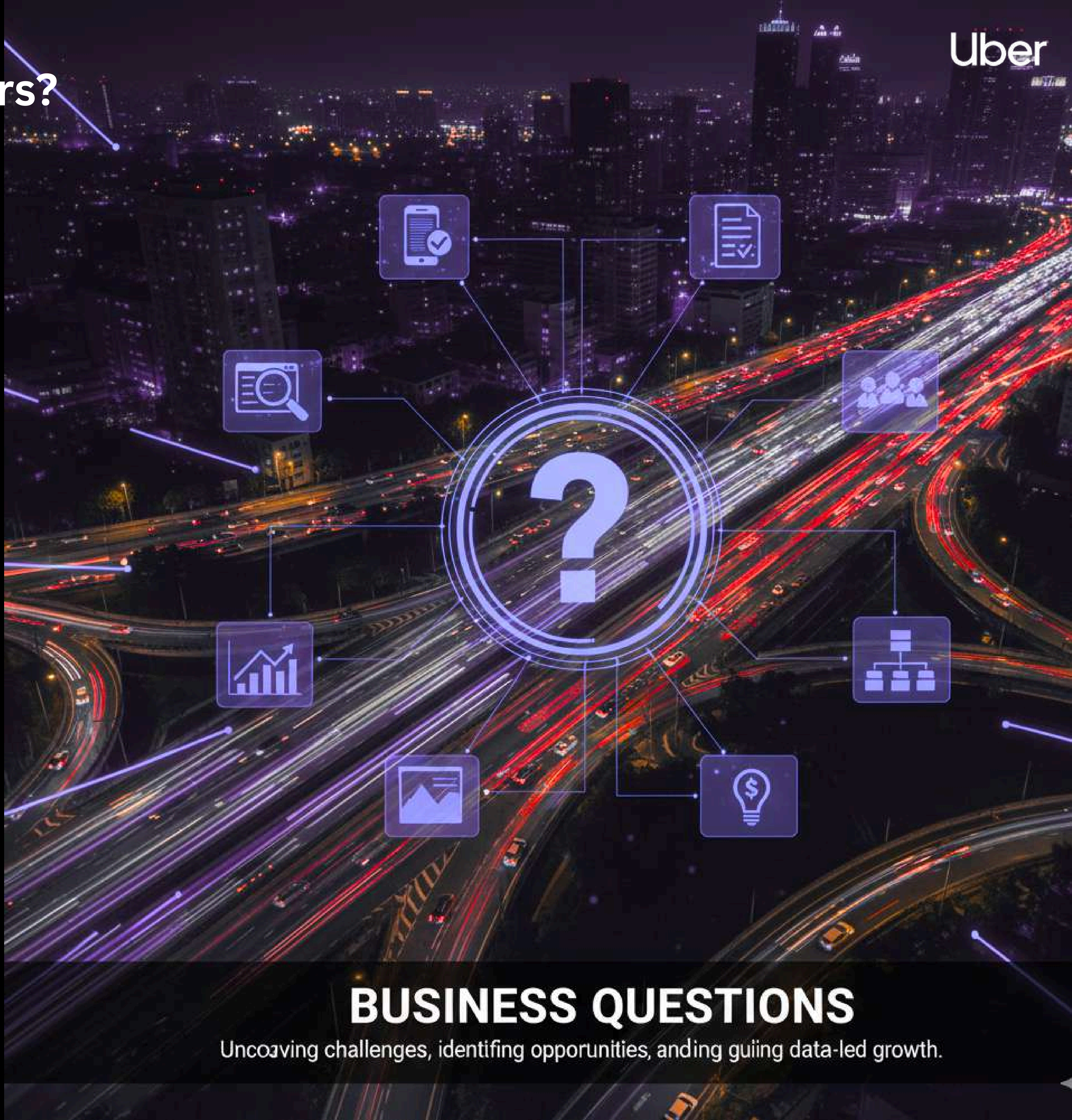
BUSINESS QUESTION 5:
How satisfied are our customers and drivers?

The average customer rating is 4.40 out of 5, which seems healthy. However, the average driver rating is only 4.23.

This 0.17-point gap might seem small, but it's significant. It tells us that while customers are generally satisfied with the service, drivers are experiencing or causing issues that negatively impact ratings.

Regular riders cancelled 11,000 trips — a major red flag showing loyal customers are walking away.

Implement driver training programs and quality incentives; close rating gap to 4.40+



BUSINESS QUESTION 6:

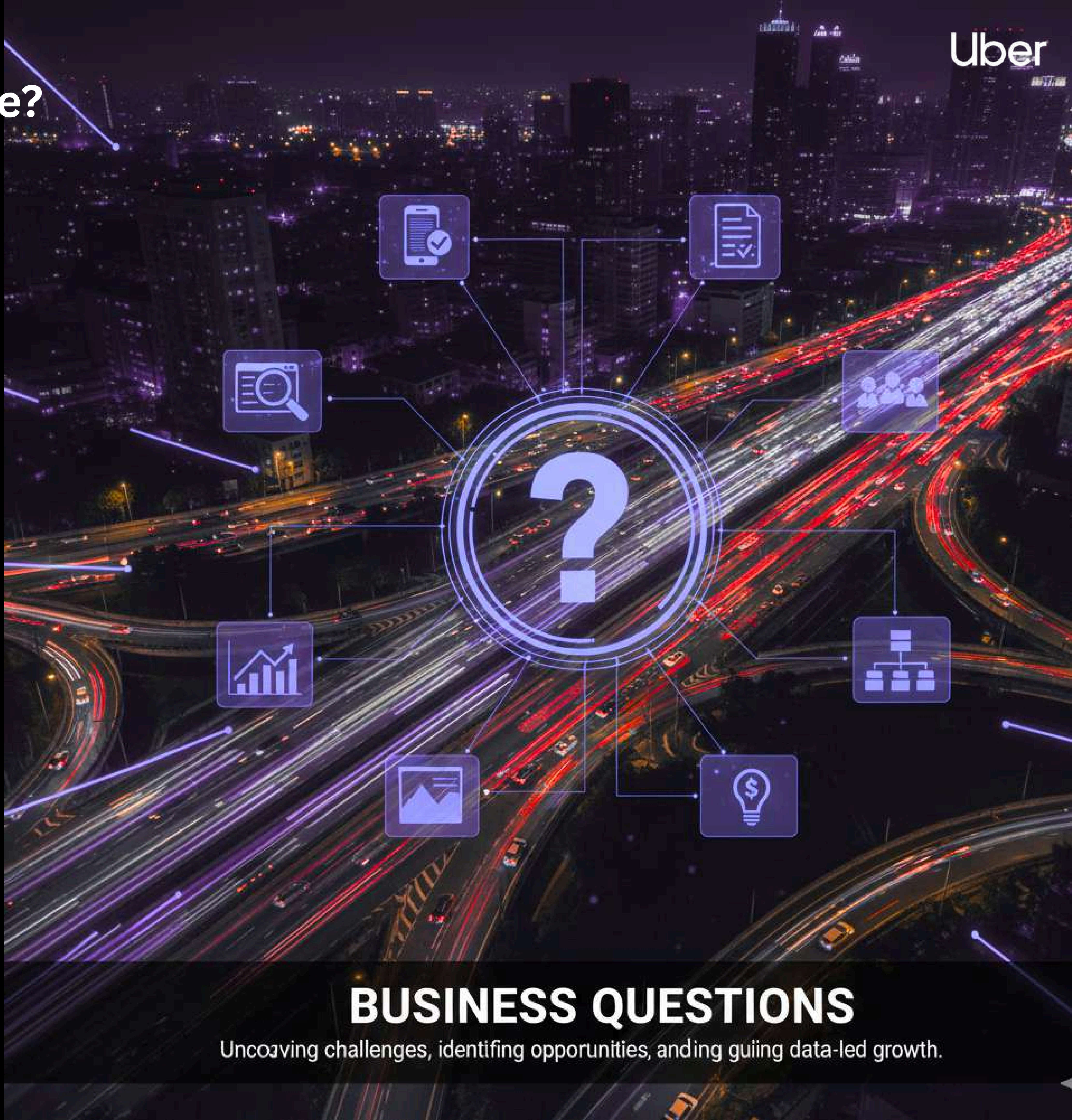
What's our customer retention looking like?

Among 104,000 active customers, true engagement varies across first-time, two-trip return riders, and regular riders with 3+ trips.

A large share of customers are one-time riders—strongly signaling an onboarding and first-impression problem.

Turning first-time riders into regulars is crucial, as acquiring new customers costs 5–7 times more than retaining them.

Create promotional offers and onboarding programs; implement loyalty rewards for regular riders



INSIGHTS

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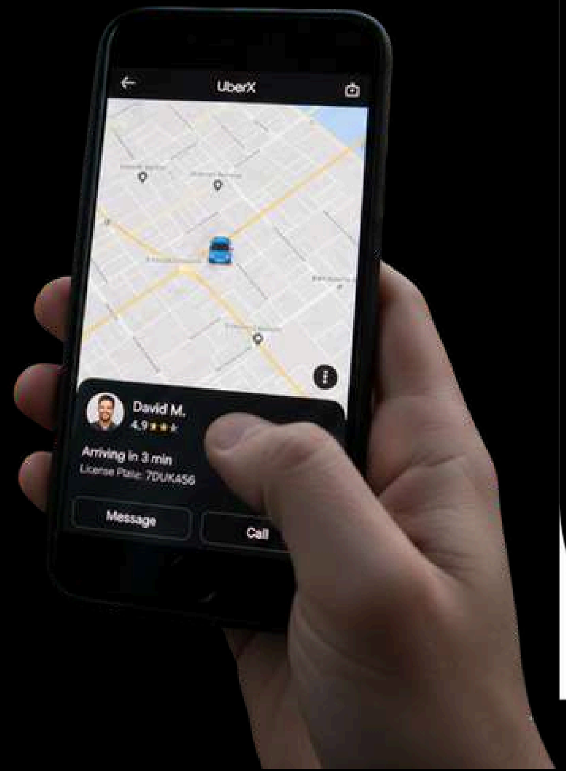
1. PERFORMANCEOF OPERATIONS

- **HIGH RATE OF COMPLETION** STRONG OPERATIONAL RELIABILITY ISINDICATED BY 93K COMPLETED BOOKINGS WITH 100% COMPLETION ACROSS VEHICLE TYPES.
- **CONSIDERABLE LOSSES :** A SIGNIFICANT REVENUE LEAK IS REPRESENTED BY 57K LOST BOOKINGS, OR ABOUT 38% OF POTENTIAL BUSINESS.
- **CONCERNING CANCELLATIONS :** 37K CANCELED RIDES INDICATE PROBLEMS WITH OPERATIONS OR CUSTOMER SERVICE THAT NEED TO BE FIXED RIGHT AWAY.



2. REVENUE PATTERNS

- **TOTAL REVENUE :** 52M GENERATED WITH ANAVERAGE FARE OF 345.64 PER TRIP
- **PAYMENT PREFERENCE :** UPI DOMINATES WITH23M (44% OF REVENUE), FOLLOWED BYCASH (13M) AND UBER WALLET (6M)
- **REVENUE VOLATILITY:** MONTHLY REVENUE FLUCTUATES BETWEEN 4.19M AND 4.42M, WITH NOTABLE DIPS IN FEBRUARY AND AUGUST
- **VEHICLE REVENUE :** AUTO GENERATES HIGHEST REVENUE (13M), FOLLOWED BY BIKE (11M) AND GO MINI/SEDAN

INSIGHTS

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3. PATTERNS OF DEMAND

- **PEAK HOURS:** ON WEEKDAYS, THEEVENING TIME SLOT (6 PM TO 9 PM) CONSISTENTLY EXHIBITS THE HIGHEST DEMAND (4,500–4,800 RIDES).
- **MODERATE MORNINGS:** 2,300–2,400 RIDES ON AVERAGE DURING THE 6–9 AM WINDOW
- **WEEKLY REGULARITY:** THROUGHOUT THE WEEK, DEMAND IS COMPARATIVELY CONSTANT WITH ONLY MINOR FLUCTUATIONS.
- **TRENDS BY SEASON:** THE DISTANCE TRAVELED FELL PRECIPITOUSLY IN FEBRUARY (197K) AFTER PEAKING IN JANUARY (232K).

GEOGRAPHICAL PERSPECTIVES

- **TOPPICKUP LOCATIONS:** THE MAIN DEMAND CENTERS ARE SAKET (931), FOLLOWED BY BARAKHAMB (846) AND KHANDSA (849).
- **TOP DROP LOCATIONS** INCLUDE LOK KALYA (916), ASHRAM (836), AND BASAI DHA (917).
- **DIFFERENT PATTERNS:** THE LOCATIONS OF PICKUP AND DROP-OFF SHOW THE DIRECTIONAL FLOW PATTERNS OF CITY TRAVEL.

INSIGHTS

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5. CUSTOMERS' BEHAVIOR

- **104K ACTIVE CUSTOMERS** WITH RATING EXTREMES (5.00 MAX, 3.00 MIN) MAKE UP THE CUSTOMER BASE.
- **SCORES FOR SATISFACTION:** THE FACT THAT THE CUSTOMER RATING (4.40) IS HIGHER THAN THE DRIVER RATING (4.23) INDICATES THAT THE DRIVER'S PERFORMANCE MAY REQUIRE IMPROVEMENT.
- **CLIENT SEGMENTS:** ALTHOUGH REGULAR RIDERS PREDOMINATE, THERE ARE SIGNIFICANT OPPORTUNITIES FOR NEW CLIENTS AS WELL AS RETURNING ONES.
- **REASONS FOR CANCELLATION:** THE TWO MAIN CAUSES OF CANCELLATIONS ARE BLANK ENTRIES (29,942) AND CUSTOMER-RELATED ISSUES (6,835).

6.VEHICLE PERFORMANCE

AUTODOMINANCE: AUTOS LEAD IN BOOKINGS (23,128),REVENUE (12.8M), AND DISTANCE (0.63M)

BIKE POPULARITY: SECOND HIGHEST WITH 20,560 COMPLETED BOOKINGS AND 11.4M REVENUE

PREMIUM UNDERUTILIZATION: PREMIER SEDAN AND UBER XL SHOW SIGNIFICANTLY LOWER USAGE DESPITE AVAILABILITY

DISTANCE EFFICIENCY: AUTO AND BIKE HANDLE SHORTER TRIPS EFFICIENTLY WHILE GO SEDAN COVERS LONGER DISTANCES (0.43M)

INSIGHTS

Uber

7. TRENDS IN GROWTH

- **GROWTH IS INCONSISTENT**, RANGING FROM -12.3% IN FEBRUARY TO +15.0% IN MARCH.
- **RECENT DECLINE:** THERE ARE ALARMING DOWNWARD TRENDS IN NOVEMBER (-1.7%) AND DECEMBER (-0.5%).
- **RECOVERY PATTERN:** OCTOBER SHOWED POSITIVE GROWTH (5.4%) AFTER NEGATIVE AUGUST-SEPTEMBER PERIOD

8.CRITICAL ISSUES

- **HIGH CANCELLATION RATE:** 11K CANCELLED RIDES BY REGULAR RIDERS INDICATES SERVICE ORMATCHING PROBLEMS
- **LOST BOOKINGS:** 57K REPRESENTS MASSIVE OPPORTUNITY COST AND COMPETITIVE VULNERABILITY
- **DRIVER-RELATED CANCELLATIONS:** "MORE THAN PERMITTED PEOPLE" AND DRIVER-SIDE ISSUES NEED POLICY ENFORCEMENT
- **DATA QUALITY:** 29,942 BLANK CANCELLATION REASONS SUGGEST INCOMPLETE TRACKING OR REPORTING GAPS

STRATEGIC RECOMMENDATIONS

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REVENUE RECOVERY & GROWTH

- **<u>REDUCE LOST BOOKINGS</u>:** IMPLEMENT SURGE PRICING OPTIMIZATION AND DRIVER INCENTIVES DURING PEAK HOURS TO CAPTURE THE 57K LOST BOOKINGS (POTENTIAL 38% REVENUE INCREASE)
- **<u>DYNAMIC PRICING STRATEGY</u>:** ADJUST FARE STRUCTURES DURING LOW-DEMAND MONTHS (FEB, AUG, NOV, DEC) TO STABILIZE REVENUE FLUCTUATIONS
- **<u>PREMIUM VEHICLE UTILIZATION</u>:** LAUNCH TARGETED PROMOTIONS FOR PREMIER SEDAN AND UBER XL TO IMPROVE THEIR CURRENTLY LOW BOOKING RATES

STRATEGIC RECOMMENDATIONS

EFFECTIVENESS OF OPERATIONS

- **PROGRAM FOR REDUCING CANCELLATIONS:** IMPROVE DRIVER-RIDER MATCHING ALGORITHMS AND PENALTY STRUCTURES TO ADDRESS THE 37K CANCELLED RIDES.
- **ENHANCEMENT OF DATA QUALITY:** IMPLEMENTING MANDATORY REASON SELECTION AT CANCELLATION WILL ADDRESS THE 29,942 BLANK CANCELLATION ENTRIES.
- **MANAGEMENT OF DRIVER CAPACITY:** DURING PEAK EVENING HOURS (6–9 PM), WHEN DEMAND REGULARLY SURPASSES 4,500 RIDES, ADD MORE DRIVERS.

STRATEGIC RECOMMENDATIONS

IMPROVEMENT OF THE CUSTOMER EXPERIENCE

- **ENHANCEMENT OF DRIVER PERFORMANCE:** CLOSE THE RATING GAP BETWEEN DRIVER RATINGS (4.23) AND CUSTOMER SATISFACTION (4.40) BY IMPLEMENTING QUALITY INCENTIVES AND TRAINING INITIATIVES.
- **RETENTION OF NEW CUSTOMERS:** CREATE PROMOTIONAL OFFERS AND ONBOARDING PROGRAMS TO TURN NEW RIDERS INTO LOYAL PATRONS.
- **FREQUENT RIDER LOYALTY INITIATIVES:** TO REDUCE ATTRITION, IMPLEMENT A REWARDS PROGRAM FOR THE HIGH-VALUE REGULAR RIDER SEGMENT.

STRATEGIC RECOMMENDATIONS

GEOGRAPHICAL OPTIMIZATION

- **STRATEGIC PLACEMENT:** DURING PEAK HOURS, PLACE MORE CARS CLOSE TO THE BEST PICKUP LOCATIONS (KHANDSA, BARAKHAMB, SAKET).
- **OPTIMIZING CORRIDORS:** ESTABLISH DEDICATED ROUTES BETWEEN PAIRS OF PICKUPS AND DROPS THAT RECEIVE A LOT OF TRAFFIC, LIKE KHANDSA TO ASHRAM.
- **POSSIBILITIES FOR EXPANSION:** EXAMINE UNDERSERVED REGIONS AND THINK ABOUT EXTENDING SERVICES TO REACH NEW MARKETS.

STRATEGIC RECOMMENDATIONS

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PAYMENT & TECHNOLOGY

- **DIGITAL PAYMENT ADOPTION:** INCENTIVIZE UPI AND WALLET USAGE (ALREADY 44% OF REVENUE) TO REDUCE CASH HANDLING COSTS AND IMPROVE TRANSACTION SPEED
- **APP IMPROVEMENTS:** ENHANCE REAL-TIME TRACKING AND ETA ACCURACY TO REDUCE CUSTOMER-INITIATED CANCELLATIONS (6,835 CASES)
- **PREDICTIVE ANALYTICS:** IMPLEMENT AI-DRIVEN DEMAND FORECASTING TO OPTIMIZE DRIVER ALLOCATION

STRATEGIC RECOMMENDATIONS

FLEET MANAGEMENT OF VEHICLES

- **FOCUS ON CARS AND BIKES:** EXPAND THE SUCCESSFUL AUTO (23,128 BOOKINGS) AND BIKE (20,560 BOOKINGS) MARKETS.
- **REBALANCING THE FLEET:** REDUCE THE NUMBER OF UNDERUTILIZED LUXURY CARS OR MOVE THEM TO AFFLUENT NEIGHBORHOODS.
- **MATCHING VEHICLE LOCATIONS:** BASED ON DISTANCE AND FARE TRENDS, MATCH VEHICLE TYPES TO PARTICULAR LOCATION REQUIREMENTS.

STRATEGIC RECOMMENDATIONS

TREND AND SEASONAL MANAGEMENT

- **HANDLE VOLATILITY** BY LOOKING INTO THE UNDERLYING REASONS FOR SHARP MONTHLY FLUCTUATIONS (-12.3% TO +15.0%) AND PUTTING STABILIZING MEASURES IN PLACE.
- **WINTER STRATEGY:** CREATE FOCUSED INITIATIVES TO STOP THE DECLINING GROWTH TREND FROM NOVEMBER TO DECEMBER.
- **COMPETITOR ANALYSIS:** DETERMINE COMPETITIVE THREATS BY EVALUATING CHANGES IN MARKET SHARE DURING MONTHS WITH NEGATIVE GROWTH.

STRATEGIC RECOMMENDATIONS

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QUALITY CONTROL & COMPLIANCE

- **POLICY ENFORCEMENT:** STRICTLY ENFORCE PASSENGER LIMITS TO ADDRESS "MORE THAN PERMITTED PEOPLE" CANCELLATIONS
- **DRIVER VERIFICATION:** IMPLEMENT REAL-TIME MONITORING TO PREVENT DRIVER-SIDE CANCELLATION ISSUES
- **SERVICE LEVEL AGREEMENTS:** ESTABLISH AND MONITOR SLAS FOR WAIT TIMES, CANCELLATION RATES, AND CUSTOMER SATISFACTION

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EXPECTED IMPACT

- 50% FEWER LOST RESERVATIONS EQUALS A POSSIBLE 19% INCREASE IN REVENUE.
- 30% FEWER CANCELLATIONS = 11K MORE RIDES COMPLETED.
- BOOST CUSTOMER LOYALTY AND RETENTION BY RAISING DRIVER RATINGS TO 4.40 OR HIGHER.
- MAINTAIN A STEADY MONTHLY GROWTH RATE OF 3–7% FOR A PREDICTABLE REVENUE TRAJECTORY.

IMPLEMENTATION PRIORITY

- **IMMEDIATE (0–3 MONTHS):** RESOLVE CANCELLATIONS, CLOSE DATA GAPS, AND MAXIMIZE CAPACITY DURING PEAK HOURS
- **SHORT-TERM (3-6MONTHS):** EXPAND DIGITAL PAYMENTS, ENHANCE DRIVER RATINGS, AND INTRODUCE LOYALTY PROGRAMS
- **LONG-TERM (6–12 MONTHS):** IMPLEMENTING ADVANCED ANALYTICS, FLEET OPTIMIZATION, AND STRATEGIC EXPANSION

Uber

THANKS FOR WATCHING

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